Loudspeaker project — three-way system D

34870 Electroacoustics

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System D (shared with Group 4): Scan-Speak 26W/8534G00 woofer box + DALI Spektor 2 (S/N 7446912 – L)

Status (branch `project-solution`). Tasks 1 and 2 are designed and simulated from the Lab D/E data; Tasks 3 and 4 are prepared (coefficients, parts list, test procedure) and wait for lab time. Every number in this report is written by the scripts (`results_task1.tex`, `results_design.tex`), so rerunning them updates the text. How to rerun everything: `Project/README.md`, section "Solution branch".

1 System and plan

The system follows Figure 1 of the guide. Amplifier channel 1 is fed by MiniDSP output 1 (digital low-pass) and drives the woofer box. Channel 2 is fed by output 2 (digital high-pass) and drives the passive board, which splits it into a low-pass for the midrange and a high-pass with an L-pad for the tweeter. Our design:

Woofer box	one vent, $L = 160\,\text{mm}$ ($f_B = 24.6\,\text{Hz}$), no extra filling
Digital crossover	LR4 at 325 Hz (two identical Butterworth biquads), woofer gain $-4.5\,\text{dB}$
Midrange low-pass	LP2: series 0.82 mH, shunt 6.8 µF
Tweeter high-pass	HP2: series 5.6 µF, shunt 0.22 mH; L-pad $R_s = 4.7\,\Omega$, $R_p = 3.3\,\Omega$
Polarity	all three units normal (+)
Acoustic crossovers	324 Hz (woofer/mid) and 3762 Hz (mid/tweeter)
Result (simulated)	150 Hz to 15 000 Hz within 3.9 dB on axis, 6.3 dB at 30° (1/6 octave)

The tools behind it, all in `Project/`:

- `matlab/export_systemD.m`: the Lab D/E data into LTspice text files and CSV.
- `ltspice/task1/`: the woofer-box model fitted to Lab D (`box_model.py`) and its LTspice circuit (`Task1_WooferBox.asc`, verified against the closed form).
- `design/xover.py`: the complete chain (biquads, passive ladders loaded by the measured Z_i, measured TFs, sum) with a search over Appendix 1 values; `design/design.json` holds the design.
- `ltspice/System_D_design.asc`: the same design in LTspice, verified against `xover.py` at all 412 measured frequencies.
- `app/`: *Crossover Studio*, an interactive version of the whole model in the browser.

2 Task 1: Low-frequency tuning of the woofer box

2.1 Thiele–Small parameters (Lab D)

The free-air and closed-box impedances were fitted together with one circuit (Lecture 7): R_E from the multimeter, S_D from the data sheet, and Bl, M_{MD}, C_{MS}, R_{MS}, L_E free. Free air loads both sides of the cone with an unbaffled piston, the closed box loads the front with a baffled piston and the back with $C_{AB} = V_B/\rho c^2$. The closed box is what pins down C_{MS} (and so V_{AS}), because S_D is fixed.

	Lab D, `analyse_labD.m`	Lab D, circuit fit	Data sheet*
f_S (Hz)	22.7	22.64	22.9
Q_{MS}	9.95	9.85	9.1
Q_{ES}	0.511	0.498	0.41
Q_{TS}	0.486	0.474	0.39
V_{AS} (L)	213	204	165
Bl (T m)	–	8.28	10.1
M_{MS} (g)	–	42.1	50.5
C_{MS} (mm/N)	–	1.174	0.95
R_E (Ω), L_E (mH)	5.7, –	5.7, 1.50	5.7, –

* f_S, Qs and V_{AS} computed from the data sheet's M_{MS}, C_{MS}, Bl, R_{MS} and $S_D = 350\,\mathrm{cm}^2$.

Our driver is softer than the data sheet (V_{AS} about 25 % larger) and has less motor (Q_{ES} 0.50 against 0.41). Both shifts point the same way as a sample-to-sample spread plus a suspension that has loosened with use. The fit reproduces both curves (Figure 1); the remaining misfit is the semi-inductive voice coil above 150 Hz, which does not matter below the crossover.

2.2 LTspice analogy circuit of the box

The circuit (Figure 2) uses the impedance analogy in all three domains, as in Lecture 7, slide 27. In the acoustic domain the cone's volume velocity $U_D = S_D u$ flows through the box network and then through the front radiation impedance. The box network is C_{AB} in parallel with a leak R_{AL} and with the vent branch $M_{AP} + R_{AP}$. The current through the vent branch is U_P, and the far field is $p \propto j\omega\rho\,(U_D - U_P)/2\pi r$. One `.step` runs the plugged box and the three measured vent lengths; a fifth step is free for trying other values. `gen_box_ltspice.py--verify` confirms LTspice equals the closed form to $2 \cdot 10^{-4}$ in every step.

Two findings from fitting the three vented impedance curves:

- **The vent does not follow the tube formula.** With $M_{AP} = \rho(L + 1.46a)/\pi a^2$ and $a = 32.5\,\mathrm{mm}$, the tuning should move from 23.4 to 19.8 Hz between $L = 160$ and 240 mm. The measurement moves half as much. We therefore fitted one effective M_{AP} per length: 67.4, 83.0, 91.2 kg/m^4 against 75.3, 89.8, 104.3 from the formula (f_B = 24.6, 22.2, 21.1 Hz). Likely causes: the length we noted is the telescope's extension, not the acoustic length, or the inner tube is narrower than 65 mm. [TODO: measure both vents' inner diameters and the telescopic range on the box.]
- **The losses come with the open vent.** With the vents plugged, the box is practically lossless (the closed-box peak, 121 Ω, needs no leak). With a vent open, the curves need $Q_L = 5.2$ and $Q_P = 30$. That is the signature of port losses, not of a leaky box, so the plugged and vented configurations keep separate losses.

2.3 Validation against the near-field measurements

Figure 3 compares the near field predicted from the impedance fit, $p \approx j\omega\rho U/\pi a$ at the centre of each opening, with the measured cone and vent pressures in Pa/V. Nothing in this figure is fitted. Levels agree within 2–3 dB, the vent and cone curves cross where measured, and the cone notch falls 1 Hz to 2 Hz above the measured one. Above about 60 Hz the vent microphone picks up the cone, so the measured vent curve stops falling there. Below 120 Hz the room modes of room 026 add ripple the model does not have.

2.4 Chosen tuning

Figure 4 and the table compare the options. Because Q_{TS} is high (0.47) and V_{AS} much larger than V_B ($\alpha = 2.3$), no standard alignment from Lecture 8 can be reached. More importantly, f_3 hardly depends on the tuning: every f_B from 22 to 30 Hz gives $f_3 \approx 33$ Hz, and only the ripple grows.

Configuration	f_B (Hz)	f_3 (Hz)	ripple (dB)
Vents plugged (closed)	–	37.4	+0.7
One vent, $L = 160$ mm	24.6	32.9	+1.8
One vent, $L = 200$ mm	22.2	33.4	+1.5
One vent, $L = 240$ mm	21.1	33.8	+1.4
$M_{AP} = 52$ (e.g. both vents)	28.0	32.7	+2.4

f_3 against the mean level at 150 Hz to 300 Hz, half space, no room; ripple = highest point below 150 Hz.

Choice: one vent at $L = 160$ mm. It extends f_3 from 37 to 33 Hz at 1.8 dB ripple. Cone excursion between 25 and 40 Hz drops to about a third of the closed box's (Figure 4, bottom), so the woofer plays louder before it distorts. It is also a configuration we actually measured, so the near field validates it directly. Opening both vents would raise f_B to about 28 Hz for no gain in f_3 and +2.4 dB ripple. Filling would add a few hertz of extension but is not modelled reliably: we have no measurement with known filling. A listening room adds its own low-frequency gain, so a mild hump of under 2 dB in half space is on the safe side.

3 Task 2: Modelling the crossover filters

3.1 Measured data in LTspice

`export_systemD.m` runs `Matlab2LTspice.m` on Z_i of the midrange and tweeter (no distance) and on the Lab E p/V of all three units, on axis and at 30°, all with $d = -1.15$ m. It then gives every generator and label a unit suffix (`Vspkr_mid`, `Pout_tw`, ...). The three Lab E units were measured in one run, so their relative phase is real, and the same d is applied to all of them. `ltspice/units/Unit_*.asc` are the course template with our files, driven straight from the source. LTspice reproduces the measured $|Z_i|$ and p/V to 0.04 dB and 0° at its own frequency points (`gen_templates.py--verify`). The tweeter Z_i starts at 99 Hz and its p/V is noise below about 120 Hz, which is irrelevant behind a high-pass.

3.2 Choosing the crossover frequencies

- **Woofer–midrange.** The 26 cm woofer ($a = 10.5$ cm) reaches $ka = 1$ at 520 Hz and breaks up above 1.5 kHz, where its 30° curve falls apart. The midrange has its resonance at 90 Hz and is flat from 150 Hz. Hence 300 Hz to 400 Hz.
- **Midrange–tweeter.** The tweeter's resonance is at 1.45 kHz (Lab D) and it is about 10 dB more sensitive than the midrange. The midrange is still flat at 30° up to about 4 kHz. Hence 2.5 kHz to 4 kHz, with an L-pad.

3.3 Design method

`design/xover.py` computes, for 1 V at the MiniDSP input,

$$p = p_w\, g_w H_{LP}(f)\, TF_w(f)\, B(f) + p_m H_{HP}(f)\, T_m(f)\, TF_m(f) + p_t H_{HP}(f)\, T_t(f)\, TF_t(f),$$

where H_{LP} and H_{HP} are the MiniDSP biquads evaluated at $e^{j\omega T}$, $f_s = 96$ kHz (the strings are identical to `DigitalCrossover.m`'s, checked in MATLAB). T_m and T_t are the passive ladders, solved with ABCD matrices and loaded by the measured Z_i. TF is the Lab E data and B is the Task 1 box tuning (vented over plugged, from the fitted model), because Lab E had the vent plugged. The design was found by coordinate descent over the allowed choices: f_c in 25 Hz steps, BW2/LR4/BW4, woofer gain in 0.5 dB steps, every passive value from Appendix 1, L-pad values and polarities. The cost function combines:

- the RMS deviation of the 1/6-octave-smoothed sum from flat, 150 Hz to 15 000 Hz, on axis plus half the 30° value;
- a penalty for cancellation, where the sum falls below the power sum;
- penalties keeping the acoustic crossovers inside 250 Hz to 600 Hz and 2.2 kHz to 4 kHz, and the tweeter 12 dB down at its resonance.

Without those penalties the search pulled the mid/tweeter crossover down to 2 kHz. That is flat on paper and wrong for the tweeter.

3.4 Digital crossover woofer–midrange

LR4 at 325 Hz: two identical Butterworth biquads ($Q = 0.707$) per output. `DigitalCrossover.m` only gives Butterworth ($n = 4$ gives $Q = 0.541$ and 1.307), so the LR4 is made by running it with $n = 2$ and using that section twice. LR4 won over BW4 and BW2 because the two outputs stay in phase through the crossover (Figure 6). The woofer is −4.5 dB relative to the mid/tweeter channel, which is the MiniDSP output 1 gain. Section 4.1 lists the coefficients.

3.5 Passive crossover midrange–tweeter

- **Midrange:** second-order low-pass, series 0.82 mH, shunt 6.8 µF($1/2\pi\sqrt{LC} = 2131$ Hz). The driver's rising impedance (8.5 Ω at 2 kHz, 13.6 Ω at 5 kHz) makes the filter peak slightly, which fills the midrange's own roll-off. That is why the electrical corner sits below the acoustic crossover.
- **Tweeter:** second-order high-pass, series 5.6 µF, shunt 0.22 mH($1/2\pi\sqrt{LC} = 4534$ Hz). The L-pad ($R_s = 4.7\,\Omega$ in series, $R_p = 3.3\,\Omega$ across the tweeter) takes about −11.1 dB off a 4 Ω tweeter and gives the filter a more resistive load around the tweeter's 1.45 kHz resonance.
- The acoustic slopes (filter × driver) cross at 3762 Hz in phase with normal polarity. With second-order electrical filters this is not the textbook "invert the tweeter" case, because the drivers' own roll-offs and the offset between the acoustic centres add their own phase.
- Amplifier channel 2 never sees less than 4.4 Ω (Figure 7).

3.6 Combined system response

Figure 5 shows the three contributions and the sum. On axis the 150 Hz to 15 000 Hz response stays within 3.9 dB (standard deviation 0.90 dB); at 30° within 6.3 dB. The sum is 3 dB above the power sum exactly at both crossovers and never below it (Figure 6): no cancellation anywhere. The summing method is the one in the guide, series E sources with polarity factors, in `System_D_design.asc` (Figure 9). LTspice and `xover.py` agree to < 0.01 dB at every measured frequency.

Tolerances. A Monte Carlo of 400 runs varied the coils by ±5 %, the capacitors by ±10 % and the resistors by ±5 %. The 95 % band of the sum is at most 1.7 dB wide, and coil resistance of 0.3–0.6 /mH changes it by at most 0.45 dB (Figure 8). The design does not depend on lucky part values.

4 Task 3: Implementing the crossovers

4.1 MiniDSP configuration

Routing Input1→Output1 and Input1→Output2; on both outputs only Xover active (PEQ, Comp, FIR bypassed). Paste into Output 1 → Xover → Advanced, then Process (file `data/dsp/MiniDSP_woofer_out1.txt`):

```
biquad1,
b0 = 0.000111435745339,
b1 = 0.000222871490678,
b2 = 0.000111435745339,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
biquad2,
b0 = 0.000111435745339,
b1 = 0.000222871490678,
b2 = 0.000111435745339,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
```

and into Output 2 (`data/dsp/MiniDSP_midtw_out2.txt`):

```
biquad1,
b0 = 0.985071534815898,
b1 = -1.970143069631796,
b2 = 0.985071534815898,
a1 = 1.969920198141118,
```

```
a2 = -0.970365941122475,
biquad2,
b0 = 0.985071534815898,
b1 = -1.970143069631796,
b2 = 0.985071534815898,
a1 = 1.969920198141118,
a2 = -0.970365941122475,
```

Output 1 gain −4.5 dB, output 2 gain 0 dB. The LTspice strings for the same filters are in `data/dsp/LTspice_*.txt`.

4.2 Passive filter board

Branch	Part	Value	Source
Midrange	series inductor	0.82 mH	course basket (Appendix 1)
Midrange	shunt capacitor	6.8 µF	course basket
Tweeter	series capacitor	5.6 µF	course basket
Tweeter	shunt inductor	0.22 mH	course basket
Tweeter	L-pad series resistor	4.7 Ω	course basket
Tweeter	L-pad shunt resistor (across the tweeter)	3.3 Ω	course basket

All values are single standard parts. Wiring order from the amplifier: tweeter branch $C \rightarrow$ node with L to ground $\rightarrow R_s \rightarrow$ tweeter with R_p across it; midrange branch $L \rightarrow$ midrange with C across it. Both branches start from amplifier channel 2, and both drivers are wired with normal polarity. [TODO: photo of the board, actual parts used, measured values if an LCR meter is available.]

4.3 Anechoic measurement of the complete system

Plan for the session: vents covered (the chamber is useless below 125 Hz anyway).

1. Each unit alone through its full chain (MiniDSP active, other outputs muted): compare with the dashed unit curves of Figure 5.
2. All three together, 0° and 30°: compare with the sum.
3. Woofer polarity flipped: there must be a deep notch at 324 Hz. Tweeter flipped: a notch at 3762 Hz. This checks the wiring and the phase model in one go.

[TODO: measured vs. simulated, and what we changed.]

5 Task 4: Listening test

Procedure (IEC 60268-13 room, Lecture 10), planned:

- **Set-up:** the box on the floor at the standard position, the listener at 2.5 m on axis at ear height near the tweeter. Level matched to 80 dB(A) on pink noise with a sound level meter, so loudness does not decide the comparison.
- **Comparison:** paired, A/B. A = the design, B = a variant (woofer gain ±2 dB, tweeter R_s one value up or down, vent plugged vs. open). The operator switches; the listener does not know which is which. Each pair is repeated with the order reversed.
- **Material:** short excerpts (20–30 s) with a male voice, acoustic guitar/piano, a track with steady bass, and pink noise.
- **Questionnaire:** for each excerpt, preference A/B/none, and a 1–5 scale for bass quantity, midrange naturalness (voices), treble brightness, and overall. Free comments.
- **Subjects:** at least 6, including people outside the group, with a short training round.

[TODO: results, analysis (count preferences, mean scores per attribute) and the iteration it led to.]

6 Conclusion

[TODO: after Tasks 3 and 4: final design, measured vs. simulated, what we would change.]

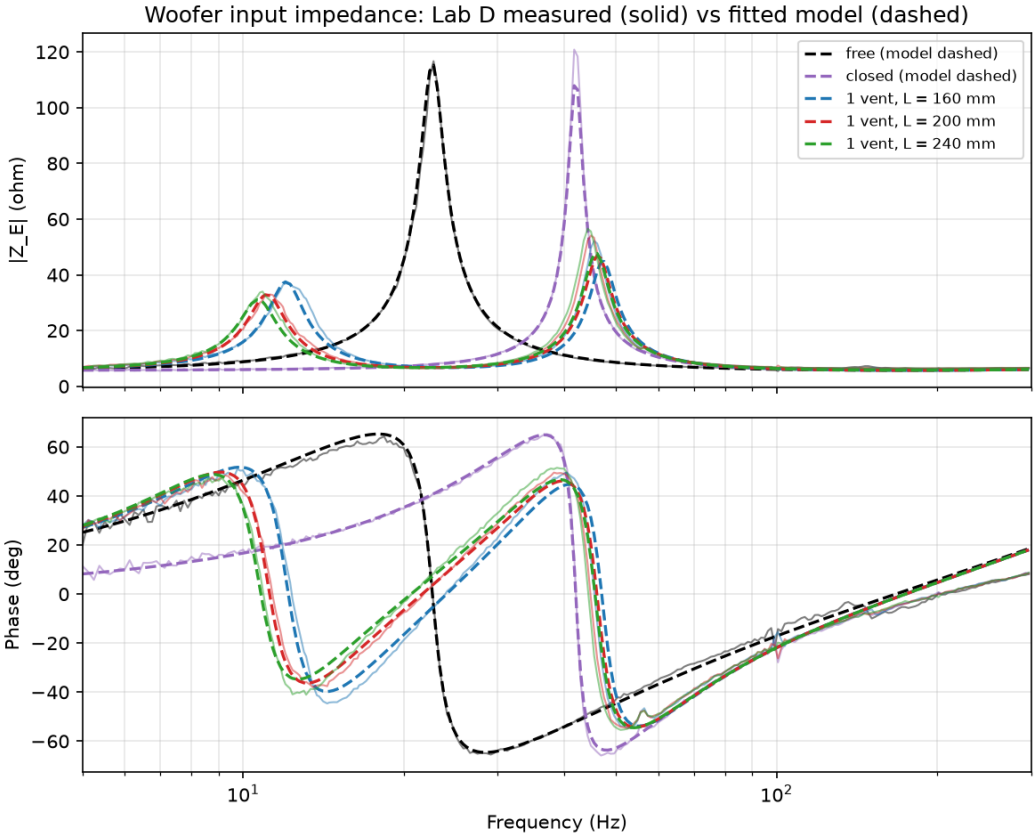


Figure 1: Woofer input impedance, Lab D measured (thin) and the fitted circuit (dashed): free air, closed box and one vent at three lengths.

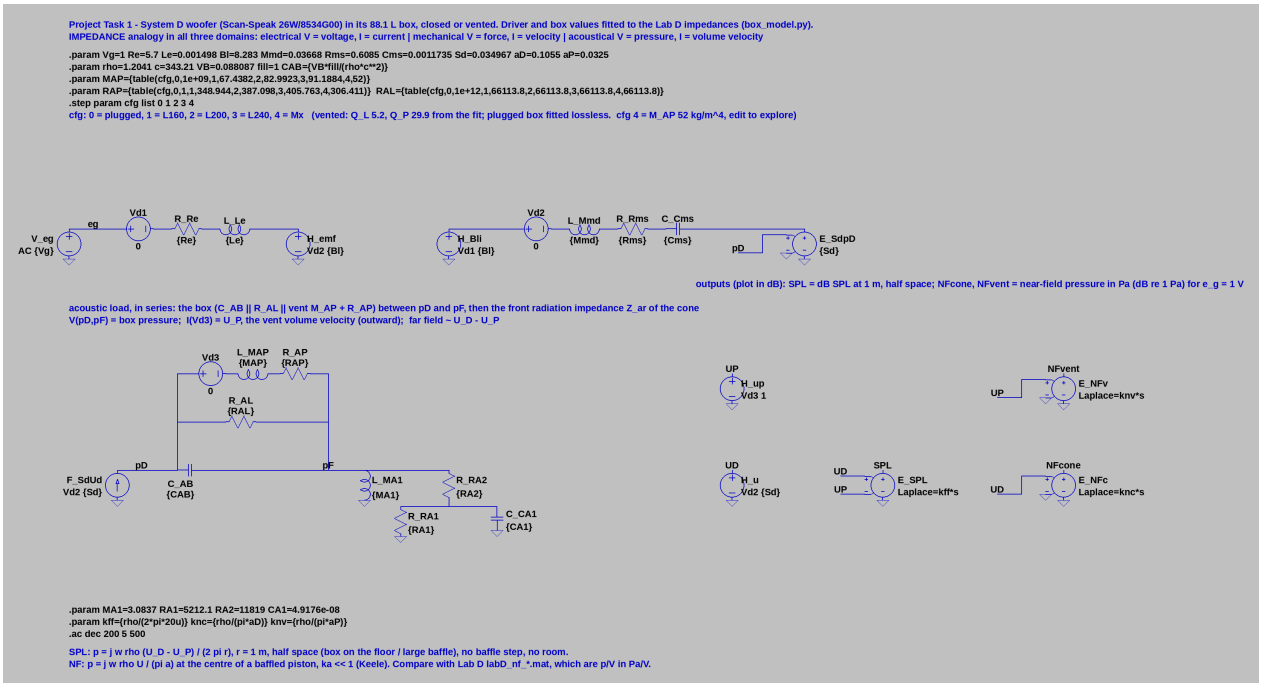


Figure 2: `Task1_WooferBox.asc`: woofer, box and vent as an impedance analogy (LTspice).

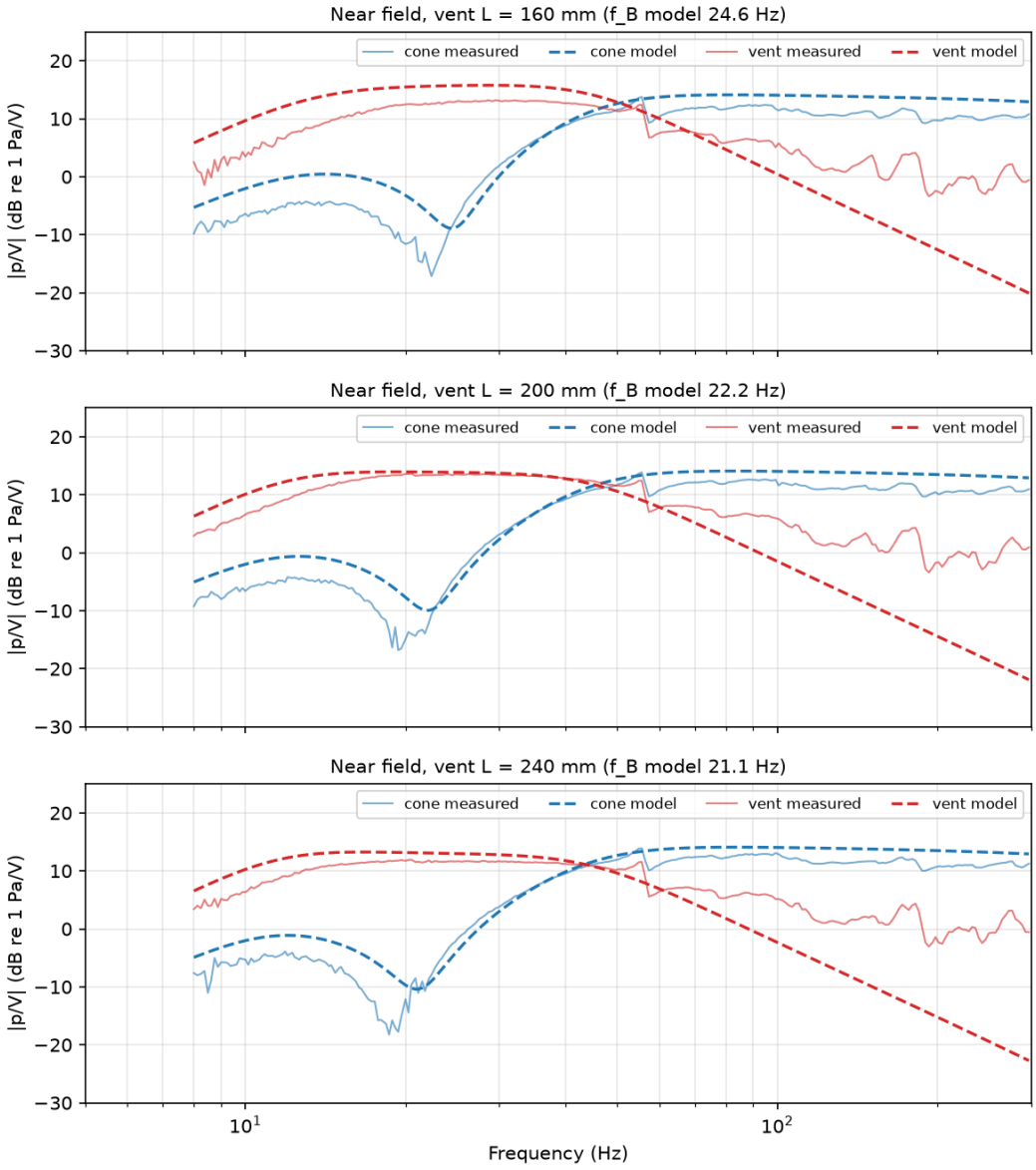


Figure 3: Near field of cone and vent, measured (Lab D) and predicted by the circuit. Not fitted.

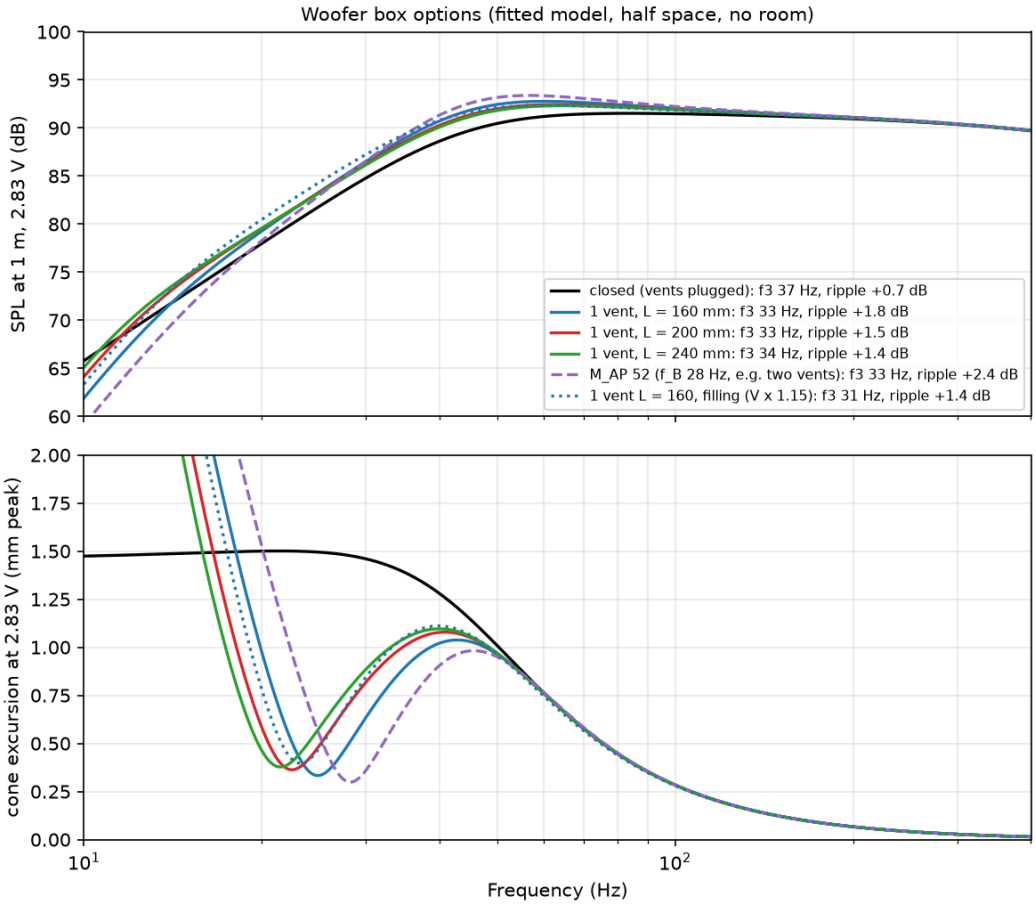


Figure 4: Box options: response at 1 m (half space, 2.83 V) and cone excursion.

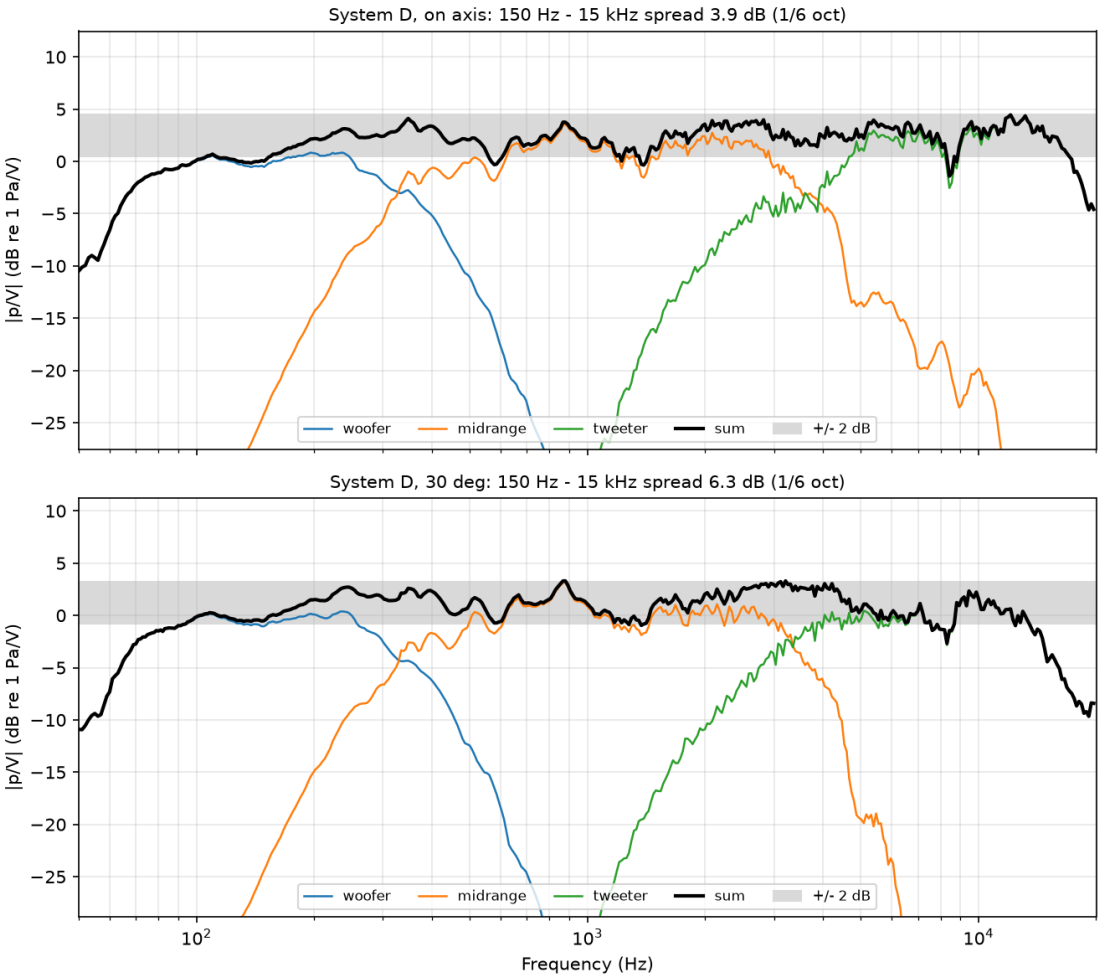


Figure 5: System response from the measured data: woofer, midrange, tweeter and sum, on axis and at 30°.

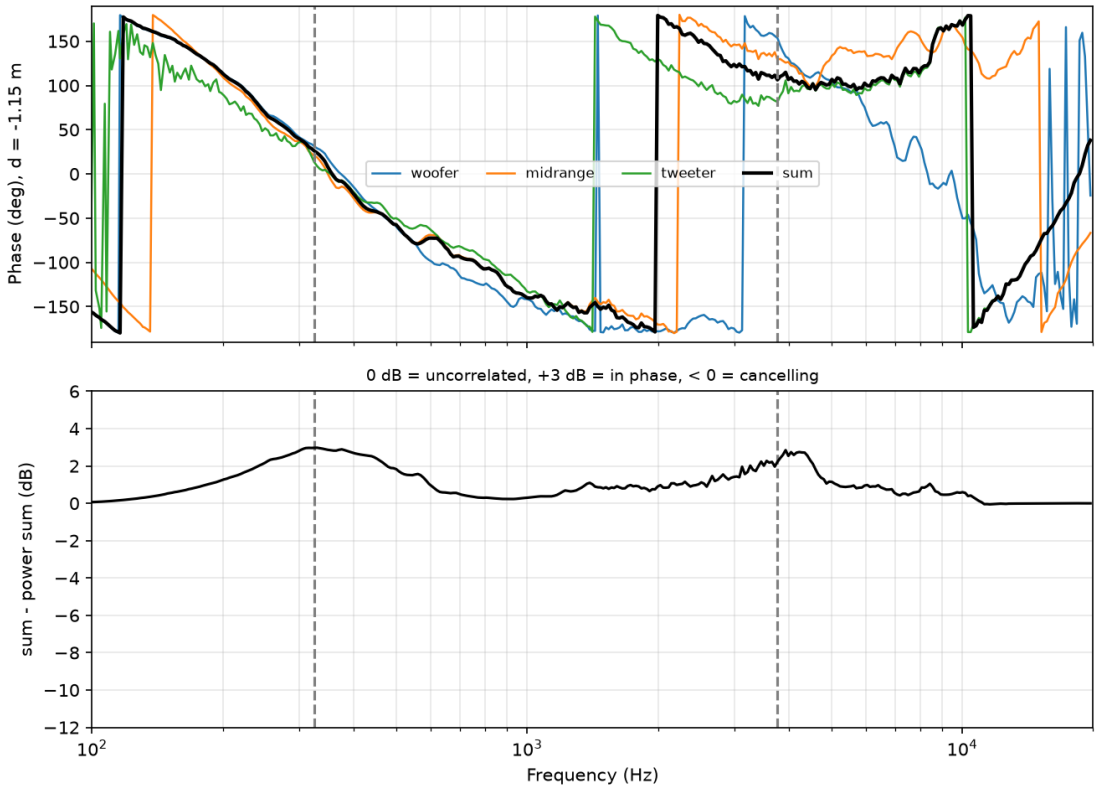


Figure 6: Phase of the three contributions ($d = -1.15\,\mathrm{m}$) and the sum relative to the power sum: in phase at both crossovers.

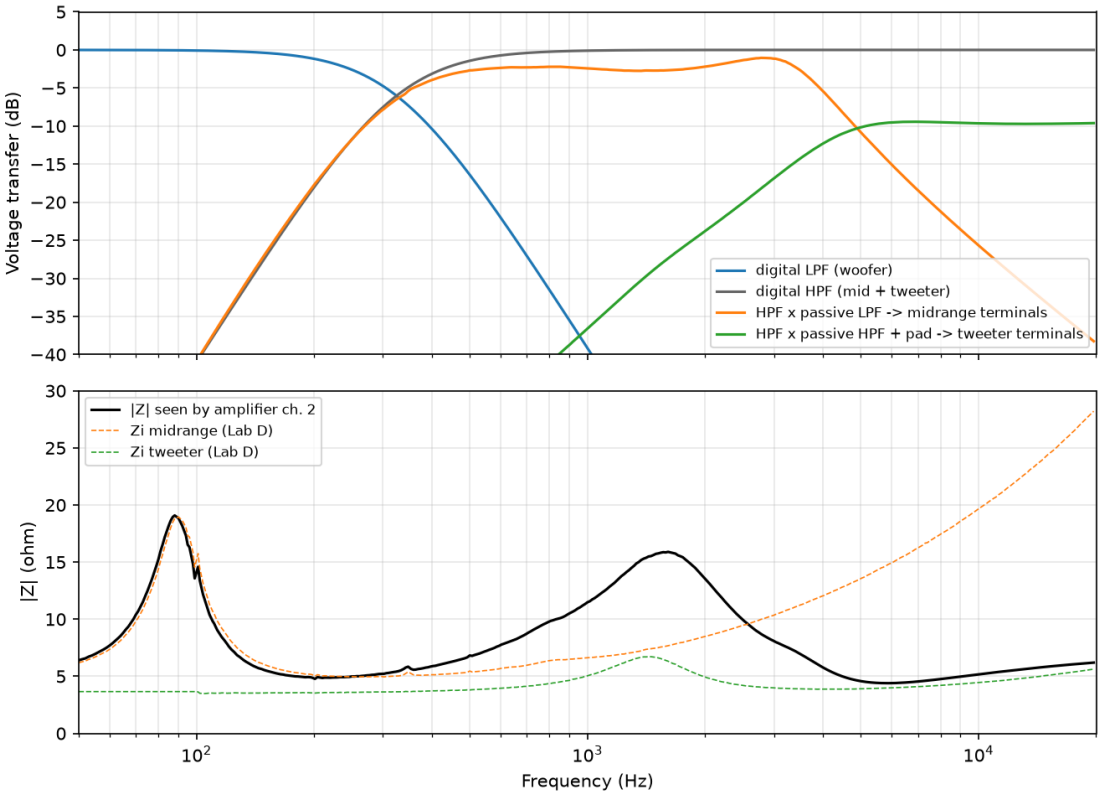


Figure 7: Electrical transfer of the digital and passive filters at the driver terminals, and the load on amplifier channel 2.

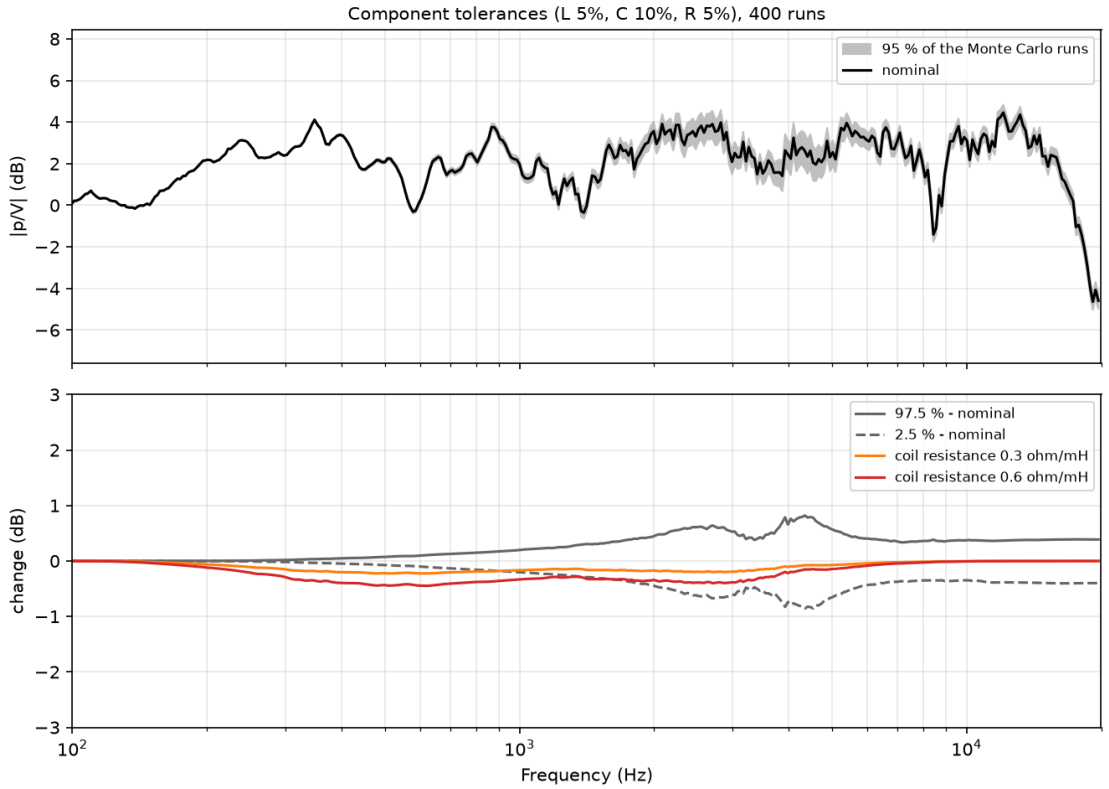


Figure 8: Component tolerances and coil resistance.

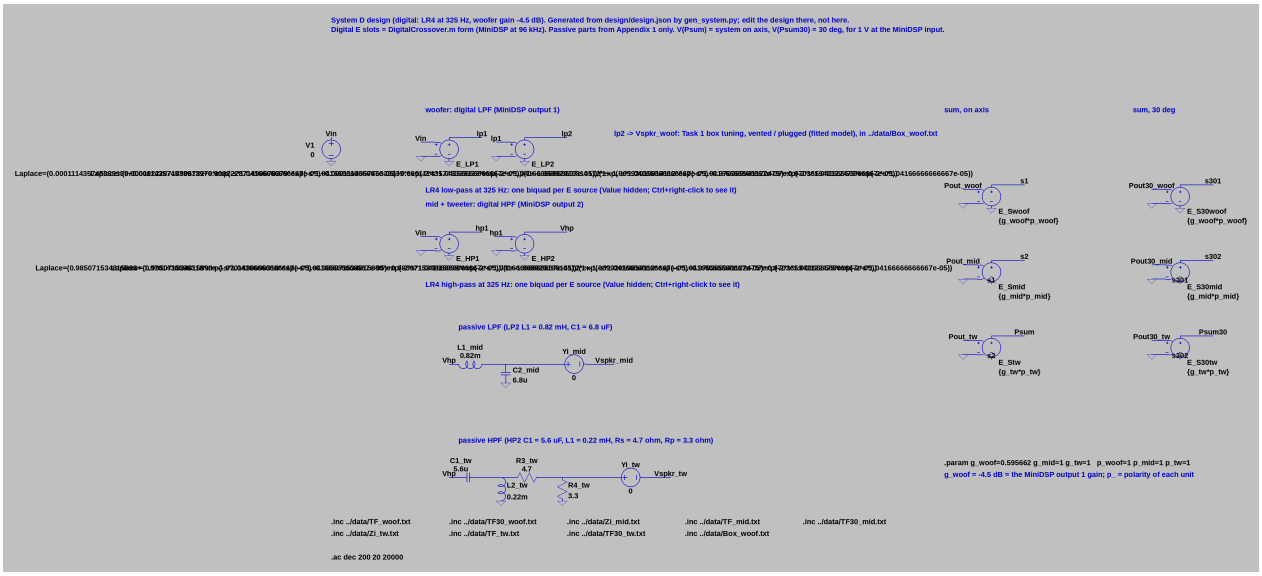


Figure 9: `System_D_design.asc`: the complete system in LTspice (the long biquad strings are hidden in LTspice; this rendering shows them).